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TDM Rights Reservations and Third-Party Republication in Common Crawl: An Empirical Study

Bachelor's Thesis

Tudor Pitulice, 234803, 234803@buas.nl
Supervisors: Mekselina Doganc & Peiman Barnaghi

Abstract: Large language models are largely pretrained on text from the online web archive: Common Crawl. Under EU copyright law, publishers may reserve their content from text and data mining by machine-readable means, and robots.txt is one prominent mechanism recognised in the EU Code of Practice for General-Purpose AI Models. Such a file applies only to the host serving it, and whether reserved content reaches the archive through copies on other websites has not been measured. For Nature, The Guardian and The New York Times, 3,622 pre-2018 articles were searched for in two complete Common Crawl snapshots, crawled in December 2018, before any reservation, and in January 2026, after all three publishers had made one. A three-stage pipeline combined word-sequence screening, embedding-based retrieval and character-level containment; candidate passage pairs above the threshold achieved an estimated precision of 0.95 (95% CI 0.83–0.99) on a manually labelled sample. Articles later subject to the publishers' reservations were found on third-party websites in both snapshots for all three publishers (2,686 occurrences in 2018, 776 in 2026). Occurrences on the publishers' own domains fell from 603 to none, while third-party occurrences per reference article declined by 53–88% but did not disappear. These results show a technical limit of host-level signalling and detection in a multi-host archive: the reservation on the original host is not visible on other hosts carrying the same text. Absence from the publisher's own domain is therefore an incomplete measure of an opt-out; the reservation's legal effect on such copies is not determined here.

1 Introduction

Large language models (LLMs) are pretrained on text corpora collected at web scale, much of which is gathered by external actors rather than by the model developers themselves. A primary source of this data is Common Crawl, a non-profit organisation that has published crawls of the public web since 2008 (Common Crawl Foundation, nd), supplies the raw material for a large share of publicly documented pretraining corpora (Baack, 2024). In the mixture used to train GPT-3, a filtered version of Common Crawl was the largest single dataset, contributing 410 billion tokens and 60% of the examples drawn during training (Brown et al., 2020). Widely used derived corpora, among them C4, OSCAR, RefinedWeb and FineWeb (Raffel et al., 2020; Abadji et al., 2022; Penedo et al., 2023, 2024), are each produced by applying a different cleaning and deduplication pipeline to the same underlying archive. Therefore, Common Crawl occupies a position that no individual developer does. Rather than functioning as one dataset among many, it serves as the foundational layer from which numerous datasets are constructed. A page that enters a snapshot is ingested by every pipeline that builds on that snapshot, and its inclusion in any given derived corpus is determined by the filtering criteria of each downstream pipeline rather than by the original publisher (Baack, 2024). Because the documented construction proce-

dures for widely used derived corpora do not include a dedicated mechanism for detecting and excluding content subject to a publisher's reservation of rights, the presence of such material in Common Crawl can propagate into multiple downstream datasets (Raffel et al., 2020; Abadji et al., 2022; Penedo et al., 2023, 2024).

Regulation (EU) 2024/1689, the AI Act, does not specify which content may be used to train a general-purpose AI model. Article 53(1)(c) instead requires each provider to put in place its own policy for complying with Union copyright law, and in particular to identify and comply with reservations of rights expressed under Article 4(3) of Directive (EU) 2019/790 (European Parliament and Council of the European Union, 2019, 2024). The Code of Practice for General-Purpose AI Models, published by the European Commission in July 2025, sets out how providers may demonstrate that compliance. Its copyright chapter commits signatories to employ crawlers that read and follow the Robot Exclusion Protocol, the robots.txt standard specified by the IETF (European Commission, AI Office, 2025a; Koster et al., 2022). Article 4 is the right to which these obligations refer. It permits reproduction and extraction of lawfully accessible works for text and data mining, but Article 4(3) makes that permission conditional on the rightholder not having reserved such use, by machine-readable means where the content is made available online (European Parliament and Council of the European Union, 2019). The regime is built on delegation where the legislator sets no list of permitted sources; the provider writes its own policy; the policy resolves to a file on a web server.

The chain ends in a file whose scope is limited to a single host: robots.txt operates on a per-host basis, governing only the server on which it is located (Koster et al., 2022). Consequently, when a publisher disallows a crawler, the reservation applies to the publisher's own host but does not extend to the same article when it is hosted elsewhere. If the article is republished on a third-party domain, the copy is governed by the robots.txt file of that new host, and the original reservation does not carry over to it.

Therefore, third-party copies fall outside the publisher's control, since the publisher cannot attach its reservation to an article once it is republished on another host. Whether the copy cites the publisher makes no difference, because a citation is addressed to human readers, whereas the reservation must be expressed by machine-readable means. A crawler that follows robots.txt consequently treats the copy as unreserved, and once the copy enters Common Crawl it becomes available to every training corpus built from the archive, as described above. A provider can thus follow every robots.txt file it encounters and still obtain text that the publisher has reserved, while the publisher, having blocked the crawler, receives no indication that its content was collected. The remaining safeguard is therefore largely complaint-based: a rightholder can request the removal of a copy only after discovering its presence, and copies that remain undiscovered are not identified by this mechanism. Whether such copies occur, and on what scale, has not yet been measured previously (Baack, 2024).

This study therefore addresses the following research question:

> *RQ. Can a publisher's content appear in Common Crawl despite a machine-readable reservation of text and data mining rights, and does its observable presence differ between a snapshot crawled before such reservations were made and one crawled after?*

To answer it, three publishers that disallow Common Crawl's crawler in their robots.txt were selected, a reference cor-

pus was assembled from their articles, and two snapshots of the archive, crawled in 2018 and 2026, were searched for text from those articles. The question is operationalised as two hypotheses:

- ***H1.*** At least one occurrence of opted-out content is present in any Common Crawl snapshot examined.
- ***H2.*** The number of occurrences per publisher differs between the two snapshots.

The two hypotheses correspond to the two parts of the research question, with H2 building on H1. H1 establishes whether reserved content appears in the archive at all, for which a single occurrence is sufficient. H2 then asks whether the extent of that presence differs between a snapshot crawled before such reservations were made and one crawled after. H1 is therefore a precondition for H2 rather than a separate test: without detected occurrences there would be no difference between the snapshots to describe, whereas H1 holds irrespective of the direction of any such difference. The two snapshots were crawled in December 2018 and January 2026, on either side of the Directive's transposition deadline of 7 June 2021 (European Parliament and Council of the European Union, 2019). H2 asks whether the two snapshots differ; it does not seek to identify the cause of any observed difference. Both snapshots were searched in full rather than sampled, so a negative result would apply to the entirety of the crawls for the selected articles from the three publishers.

Through its research question and hypotheses, this study contributes empirical evidence on whether content reserved under Article 4(3) is nevertheless present in Common Crawl. In doing so, it adds transparency to the archive from which many LLM training corpora are built, and it draws attention to a possible gap in the current European framework between the host-level mechanism through which reservations are expressed and an archive assembled from copies held on many hosts. In addition, the study provides a pipeline for searching complete Common Crawl snapshots for a fixed reference corpus, which can be applied to further snapshots and publishers without modification.

The remainder of the thesis is organised as follows. Section 2 distinguishes the mechanisms through which publishers reserve and remove content, reviews prior measurement of opt-out signals at the domain level, and identifies the absence of content-level measurement across hosts as the gap addressed in this study. Section 3 describes the reference corpus, the snapshots and the three-stage matching pipeline, together with the analysis plan. Section 4 reports the results for each publisher and evaluates both hypotheses. Section 5 discusses their implications, the limitations of the design and directions for future work, and Section 6 concludes.

2 Related Work

2.1 Mechanisms for reservation and removal

Publishers have several instruments to prevent their content from appearing in Common Crawl, and these differ both in nature and in the stage at which they take effect. Table 1 distinguishes six such mechanisms out of which one is a legal right while the remaining five are technical or procedural means through which that right is exercised. Most operate at the point of collection, when a page is crawled. Only removal requests, and the ledger that records them, act on content that has already entered the archive.

The technical mechanisms identify reserved content through the website on which it was originally published. The Directive leaves the choice of means open, but the Robots Exclusion Protocol and its crawler-specific rules are evaluated per

Table 1: Mechanisms through which publishers reserve or remove content from Common Crawl.

Mechanism	What it does	Unit	Timing	Source
Article 4(3) DSM Directive	Legal right to reserve text and data mining; for content available online, the reservation must be made by machine-readable means (Recital 18)	Legal right	Before collection	European Parliament and Council of the European Union (2019)
`robots.txt` (Robots Exclusion Protocol)	Technical signal read by a crawler when it requests pages from a host	Per host	At crawl	Koster et al. (2022)
Crawler-specific controls	Rules for a named user agent (e.g. `CCBot`, `GPTBot`, `Google-Extended`) within `robots.txt`	Per host	At crawl	Common Crawl Foundation (2026)
Removal requests	Requests to delete pages that have already been crawled	Per website	After collection	Rights Alliance (RettighedsAlliancen) (2024); Stichting BREIN (2025)
Common Crawl Opt-Out Ledger	Public list of the legal opt-out requests Common Crawl has received, naming the website owners concerned	Per website	After collection	Common Crawl Foundation (2025)
Proposed EU opt-out registry	Content-based fingerprinting to identify a work wherever a copy is held	Per work	Proposed	European Commission (2026)

host, removal requests name the websites from which content should be deleted, and the Opt-Out Ledger lists website owners, including the BBC, The Guardian, and Le Monde (Tobitt, 2026). Only the proposed registry would identify a work by its content rather than by the host on which it appears. In technical terms, `robots.txt` is evaluated at the host from which a crawler requests a page, and a rule published on one host is not consulted when the same text is served from another (Koster et al., 2022). Whether a rightholder's reservation has legal effect for copies hosted elsewhere is a separate legal question that lies outside the scope of this empirical study.

2.2 Prior measurement of opt-out adoption and its consequences

The existing empirical research on opt-outs has measured how widely restrictions are adopted. Longpre et al. (2024) audited the `robots.txt` files and terms of service of 14,000 web domains and found that, within a single year, `robots.txt` restrictions came to cover about 5% of all tokens in C4 and more than 28% of its most actively maintained, critical sources. Fletcher (2024) reported that 48% of the most widely used news websites in ten countries were blocking OpenAI's crawlers by the end of 2023, and Bouchaud and Ramaciotti (2025), extending the audit to the one million most visited websites, related the spread of such restrictions to shifts in the composition of Common Crawl and its derivative training datasets. Even though, these studies focus on various LLM training sets, the common denominator is represented by the common source which is Common Crawl. In order for the domains to reach training sets such as C4 and FineWeb, among many

other, these domains must be present first in a snapshot of Common Crawl.

Fan et al. (2025) examine the consequences of these restrictions for model performance. They apply `robots.txt` opt-outs retroactively to previously collected web data, checking each document against the `robots.txt` file of the host from which it was collected.

These studies observe restrictions where they are published, not where the content ends up. Each attributes a restriction to the domain that publishes it, so an article subject to a reservation by its publisher on an unrestricted third-party domain is counted as unrestricted, if it is counted at all. Even a retroactive check such as that of Fan et al. (2025) therefore remains a host-level test.

2.3 Content beyond the host: the measurement gap

Content, however, does not stay on the host where it was first published. Across the ClueWeb and Common Crawl collections, between 14% and 52% of the documents within a crawl are near-duplicates of other documents in the same crawl (Fröbe et al., 2021), so the same text is routinely held by several hosts at once.

Whether a publisher's content is present in the archive is a practical concern. The New York Times alleges in its complaint against OpenAI and Microsoft that Common Crawl contained at least 16 million unique records of its content (The New York Times Company, 2023), and the publishers suing Google in 2026 likewise allege that Google used web-scraped datasets derived from Common Crawl (Hachette Book Group, Inc. et al., 2026). The instruments used in response, however, target the publishers' own websites. After a request from the Danish Rights Alliance, Common Crawl stopped collecting content from the websites of a number of Danish media houses and agreed to review its existing datasets (Rights Alliance (RettighedsAlliancen), 2024), and following a request by BREIN it agreed to remove more than two million articles published on Dutch news websites (Stichting BREIN, 2025). Such removal has been reported to proceed slowly and partially, with publishers receiving progress reports of 50%, 70% and 80% (Reisner, 2025), and a single Ledger entry submitted by the News/Media Alliance covers more than 900 news websites (Tobitt, 2026). In each case, the request names the websites on which the content was first published, so a copy of the same article held on another host falls outside it.

The European Commission's feasibility study on an EU-level registry of text and data mining opt-outs responds to this limitation by examining content-based fingerprinting, which recognises a work from its content regardless of where a copy is held (European Commission, 2026). As Gentile (2026) notes, digital copies replicate, lose their metadata and resurface elsewhere on the web, beyond the rightholder's control. Such a mechanism remains a proposal, however, and the extent of the problem it targets has not been measured.

The closest evidence of reserved content travelling through copies comes from outside training data. In tests of eight AI search engines, Jaźwińska and Chandrasekar (2025) observed that publishers who had opted out of crawling still saw their articles surface in results through syndicated or unauthorised copies, attributed to the sites that had republished them. This route has thus been observed in search outputs, but not measured in a training archive.

Prior work measures the adoption and consequences of opt-out signals at the domain or host level. The current study instead contributes content-level detection across hosts, testing whether selected reserved publisher content can still be detected in Common Crawl when it is served from domains other than the publisher's

own, and comparing this presence between a snapshot crawled before the reservations and one crawled after.

3 Methodology

3.1 Design Overview

This study is built on the comparison of two text corpora. The first is a reference corpus of passages drawn from three publishers: The Guardian, The New York Times, and Nature, each of which uses its robots.txt file to disallow Common Crawl's crawler, `CCBot`, from entering its official domain. The second is a target corpus consisting of a complete Common Crawl snapshot: the plaintext of every page the crawler captured during a single monthly crawl. The question the design answers is whether passages from the reference publishers are present in the target corpus on Common Crawl domains including the publishers' own.

Answering that question at the scale of a full snapshot requires more than one test, because no single test is both affordable at that scale and precise enough to identify a match. The design applies three stages in sequence, each more expensive per page than the preceding stage and applied to a progressively smaller set of pages. A lexical screen based on word 8-grams is applied to every page in the snapshot. Surviving pages are embedded and retrieved against the reference corpus by cosine similarity. Retrieved pairs are then scored by character-level containment, which measures how much of the shorter passage's text appears in the longer one. A pair is treated as a match only when it satisfies both semantic and lexical criteria. The output is a set of pairs, each linking a URL in the snapshot to a specific reference article, aggregated by domain.

3.2 Publisher selection

The three reference publishers were selected according to two criteria. First, each must disallow `CCBot` in its robots.txt, since a publisher that has not blocked the crawler provides no opt-out for the study to test. Each publisher's robots.txt was checked at both points the study compares, using archived captures rather than a single live retrieval, since a file retrieved today does not establish what it contained when either snapshot was crawled. For `nature.com`, `nytimes.com`, and `theguardian.com`, the Internet Archive capture nearest each crawl within the crawl month was retrieved and inspected manually.

In January 2026 all three carry a `User-agent: CCBot` directive followed by `Disallow: /`, applying to the entire domain. As expected, in December 2018 none of the three files contains a `CCBot` user-agent group at all, which restricts a set of non-article paths and leaves article content crawlable. The selection criterion is therefore satisfied at the time of the later crawl, and the earlier crawl predates the reservation in all three cases. The individual captures are listed in Appendix A.

Second, each must expose article text through a public API, so that the reference corpus can be assembled without crawling the publishers' own sites. As scraping a domain that disallows automated collection would be inconsistent with the premise of the study. The three publishers span news, general reporting, and academic research, but the selected publishers do not constitute a representative sample of publishers that have opted out; they are the publishers that satisfied both criteria.

The three sources differ substantially in how much text they release, and this asymmetry propagates through the rest of the design. The Guardian's Open Platform API returns full article body text. The New York Times Article Search API returns no body text at all, only a headline, a lead paragraph, and a one- or two-sentence abstract, which are concatenated into a single record. Nature ab-

stracts were obtained through OpenAlex, a third-party bibliographic index, rather than from `nature.com`. OpenAlex stores abstracts as an inverted index, which is reconstructed into running text at collection time. A Guardian article therefore contributes on the order of a thousand words to the reference corpus, a Nature abstract roughly two hundred, and a New York Times record a hundred. The consequence is that the three publishers are not equally detectable: a longer reference text offers more opportunities for a republished passage to match, and a short one offers few.

Within each publisher, articles were retrieved using the rankings provided by the respective APIs rather than by random sampling. Guardian articles were drawn across nine sections, ordered by the API's relevance ranking. New York Times records were retrieved through ten broad topical queries, including the economy, climate change, technology, and elections, again ordered by relevance. Nature abstracts were drawn from three journals and ordered by citation count, so that the corpus consists of the most-cited pre-2018 articles in each. This ranking is not neutral with respect to the outcome being measured. Prominent articles are more likely to be deposited in institutional repositories, indexed by bibliographic services, and picked up by aggregators than obscure ones, which means the corpus is weighted toward exactly the material most likely to be republished. The study therefore measures the republication of prominent content, and its results should not be read as a rate applying to a publisher's output as a whole.

The endpoints, query parameters, ranking and pagination settings, retrieval dates, exclusion rules, and realized record counts for each of the three sources are recorded in Appendix B, together with the corpus manifest that fixes the collected records independently of any later change to the APIs.

Two of the three publishers release no article body text, so the reference corpus consists largely of abstracts and lead paragraphs. An abstract condenses a paper's problem, method, and result into a single passage, written so that a reader can decide from it alone whether the full paper is worth reading. Thus, it represents the paper without the supporting details. A news lead performs the same function under the inverted-pyramid convention, placing the substance of the story in its opening sentences so that a reader who stops there has still received the report. Both are written to stand in for the work as a whole, which is what makes them usable as reference passages: a match against a lead or an abstract identifies the article it belongs to, without requiring access to the body text that neither publisher releases.

All articles in the reference corpus were published on or before 31 December 2017. The cutoff is applied at corpus construction, and articles published later are discarded regardless of how they were retrieved. It exists so that every reference article predates both Common Crawl snapshots the corpus is searched against, which is what allows the same fixed set of articles to be tracked across them; the reasoning is set out with the snapshots in Section 3.3.

3.3 Reference corpus

Before the articles can be used, they undergo a series of preprocessing steps. First, each article is divided into passages, which are then embedded and compared. A whole Guardian article averaged into a single vector would be dominated by whatever the majority of its text is about, so a third party reproducing three paragraphs of it would leave a signal too weak to detect. Reference text is therefore divided into windows of four consecutive sentences. On the Guardian side these windows are non-overlapping: the first covers sentences one to four, the second

Table 2: Composition of the reference corpus. Passages are four-sentence windows; median length is measured in words per passage.

Publisher	Articles	Passages	Pass. per article	Median words
Guardian	220	2,040	9.27	96
NYT	2,404	2,449	1.02	32
Nature	998	2,427	2.43	95
Total	3,622	6,916	1.91	—

sentences five to eight, and so on. Where the sentence count is not a multiple of four, a final window covering the last four sentences is emitted, so that the closing sentences of an article are never dropped even though this window partly repeats the one before it. New York Times records are short enough that no window applies and each is embedded as a single passage.

Two additional filters are applied before a passage is kept: a passage containing fewer than fifteen words is discarded, and so is one in which letters and spaces account for less than 60% of the characters, a test that removes tables of contents, dot-leader strings, and other front matter that the bibliographic source occasionally returns in place of an abstract. Passages that pass both filters are embedded. The number of articles pulled from each publisher was set so that the three would contribute comparable numbers of passages rather than comparable numbers of articles: because a single Guardian article yields many windows while a New York Times record yields one, equal article counts would have produced a corpus dominated by Guardian text. Table 2 gives the resulting composition.

Common Crawl pages are divided into windows by the same four-sentence rule, but with a step of two rather than four, so that consecutive windows overlap by half. The asymmetry is deliberate: a reference article has a known structure and its passages can be laid out end to end without loss. A republished passage, by contrast, may begin at any point in the page on which it appears, and a non-overlapping window would split it across two windows and dilute both. Halving the step means that any four consecutive sentences on a page fall within one window or the one adjacent to it, at the cost of embedding roughly twice as many passages per page.

Because the three publishers contribute passages of very different length, and a shorter reference passage offers fewer opportunities for a match than a longer one, results are reported separately by publisher rather than aggregated across them.

3.4 Target corpus

Two snapshots were analysed: CC-MAIN-2018-51, crawled in December 2018, and CC-MAIN-2026-04, crawled in January 2026. They were chosen to sit on either side of the introduction of the EU copyright regime governing machine-readable opt-outs. Directive (EU) 2019/790 (European Parliament and Council of the European Union, 2019; Westkamp, 2024) was adopted in April 2019 and Member States were required to transpose it by 7 June 2021. The 2018 snapshot therefore predates the Directive's adoption entirely, and the 2026 snapshot postdates the transposition deadline by more than four years. Because every article in the reference corpus was published before either crawl, the same fixed set of articles can be searched for in both, and any difference in what is found is a difference in the state of the web rather than in the corpus being searched for.

The reservation under examination exists in only one of the two snapshots. At the time of the earlier crawl the Directive had not been adopted and no machine-readable reservation was available to make; at the time of the later one all three publishers had made it (Table 3) as explained in Section 3.2. The 2018 snapshot therefore records the state of the web before the mechanism existed, and serves

as a baseline rather than as a case of the mechanism failing.

Each page enters the study as Common Crawl's published plaintext rendering (Common Crawl, 2026). This extraction is mechanical and does not separate an article from the page around it, so navigation menus, related-link lists, and footers appear in the text alongside the content; a dedicated extractor would produce cleaner text, but would have to run over every page in the snapshot and would introduce a second extraction step, increasing the computational requirements of the pipeline. Using the published plaintext directly means the same text is screened and verified, with no re-fetching between stages, at the cost of retaining boilerplate, which the two matching criteria are designed to tolerate rather than remove.

3.5 Scale of corpora

The reference corpus comprises 6,916 passages and the two snapshots contain 3.10 and 2.30 billion pages respectively. An exhaustive pairwise comparison is computationally not feasible at this scale: embedding every page in a snapshot would require encoding tens of billions of passages, and the great majority of that computation would establish non-matches, since almost no page in a snapshot bears any relation to a reference article. The constraint is therefore not volume alone but the combination of volume and sparsity (Panchenko et al., 2018; Fröbe et al., 2021).

The design responds to this by separating the tests according to their cost. A test cheap enough to apply to every page cannot be precise, and a test precise enough to identify a match cannot be applied to every page; the two are therefore applied in sequence rather than together. A single inexpensive test is applied exhaustively to the snapshot, and the expensive tests are applied only to what it retains. The analogy is to searching an ocean with a net rather than by inspecting every cubic metre of it: a coarse net can be drawn across the whole of it, and although it retains a large quantity of irrelevant material, its mesh is fine enough that the target does not pass through, and what it retains is small enough to examine in later stages.

The screen retained 779,075 of the 3.10 billion pages in the 2018 snapshot and 514,219 of the 2.30 billion in the 2026 snapshot, approximately 0.02% in both cases. The later stages are therefore applied to roughly one page in every four thousand, which is what renders exhaustive coverage of a snapshot tractable. The proportion depends on how the screen is parameterized as well as on how rare the material is; Section 3.6 sets out how that parameter was chosen, and the figure reported here is the one it produced.

3.6 Stage 1: Lexical screening

The first stage operates on word 8-grams where every sequence of eight consecutive words in the reference corpus is compiled into a pattern set, 413,105 distinct sequences in total, and each page in the snapshot is scanned for a match against any of them. The retention criterion was held at one shared sequence. This criterion is what preserves the interpretation of the result: a page sharing no sequence of eight words with any reference passage cannot satisfy the criteria applied at the later stages, which require substantially greater overlap. Therefore, this stage was designed as a high-recall candidate filter, where the pages sharing no sequence of 8 words are discarded as topical unrelated to the publisher corpus.

The sequence length was set empirically on a subset of WET files screened before the full run. At five words, the screen retained approximately 8% of the pages it scanned, a rate that would have yielded on the order of 220 million survivors across the snapshot and would have far exceeded the capacity of the later stages; at eight words the rate fell to the

Table 3: Archived robots.txt captures used to establish crawl-time state.

	Capture date	
Publisher	CC-MAIN-2018-51	CC-MAIN-2026-04
Nature	2018-12-31	2026-01-01
The Guardian	2018-12-31	2026-01-01
NYT	2018-12-31	2026-01-01

Each capture is the Internet Archive capture nearest the crawl within the crawl month. No capture in the 2018 column contains a `CCBot` user-agent group; every capture in the 2026 column carries `User-agent: CCBot` followed by `Disallow: /`. Capture URLs are listed in Appendix A.

0.02% reported above. Longer sequences reduce the shortlist further but also raise the risk of discarding genuine republications, since a copy reproduced with minor alteration may contain no unaltered run of that length. An eight-word sequence was adopted as the shortest tested length that brought the shortlist within the capacity of the later stages.

The length filter was evaluated in terms of its effect on recall. For each page that subsequently yielded a detected occurrence, the number of distinct reference sequences recorded at screening time was retrieved. The median number of shared sequences was 33 in the December 2018 snapshot and 79 in the January 2026 snapshot, and in both snapshots 99.4% of such pages shared at least two sequences. Almost every detected occurrence was therefore retained with a substantial margin above the minimum retention criterion. The 33 pages in the earlier snapshot and the 8 pages in the later snapshot that shared exactly one sequence would have been excluded under any stricter criterion, which motivated the decision to retain pages sharing at least one sequence. The remaining source of false negatives consists of copies that have been altered densely enough to contain no unbroken run of eight words from the reference material. Thus, such copies would get discarded in the later stages of the pipeline.

However, before matching, both sides undergo an identical normalisation step. Text is lowercased, punctuation is removed, and runs of whitespace are collapsed to single spaces, so that differences in casing, quotation style, or spacing between an original and a copy do not prevent a match. The same normalisation is applied to the reference passages when the pattern set is constructed and to each page as it is scanned, which ensures that the sequences compared are drawn from a common representation rather than from two differently formatted ones.

Matching is performed with an Aho–Corasick automaton, which locates occurrences of any pattern in a set during a single pass over the text. The pattern set is compiled once and reused for every page, so the cost of scanning a page does not increase with the size of the reference corpus. Each pattern is stored with a leading and trailing space and each page is padded likewise, so that a sequence matches only at word boundaries and no match is registered on a fragment occurring inside a longer word.

For each retained page, the screen records the page's URL, the archive file it was read from, and the number of distinct reference sequences it contained, together with the page text itself. Retaining the text at this point is what allows the later stages to operate without returning

to the archive: the same extraction is used for screening and for verification, and no page is read twice.

3.7 Stage 2: Semantic retrieval

The stages apply progressively stricter criteria, and this ordering is what decides if an uncertain case is left out rather than counted in. The screen retains a page on a single shared sequence and retrieval admits any pair above a low similarity floor. Thus, neither stage represents a decision, and both are set to retain and filter rather than to judge. Judgement is made once, at the end, on material that has already survived. In a pipeline ordered this way, material retained too readily at an early stage can still be removed downstream, whereas material discarded early is unavailable to every stage that follows. Its output is therefore a conservative estimate of what is present, which is the form the hypotheses require.

In the next stage of the pipeline, each retained page is divided into passages before comparison. Sentences are grouped into windows of four, advancing two sentences at a time, so that consecutive windows overlap by half. A reference article has a known structure and can be partitioned without loss, whereas a republished passage may begin at any point on the page carrying it. On the other hand, a non-overlapping window would divide the paragraph between two windows and weaken the overall signal.

In this design, a sequence of four consecutive sentences beginning at an even position falls entirely within a single window, while one beginning at an odd position spans two adjacent windows, with each window containing three of its four sentences. This overlapping scheme ensures that no four-sentence run is represented by fewer than three sentences in any window. Without overlap, a passage could be divided arbitrarily between windows, potentially weakening the signal. Capturing every run intact at every possible alignment would require advancing by one sentence at a time, which would double the number of passages embedded from each page. A step of two therefore balances coverage against computational cost.

Passages are embedded with all-MiniLM-L6-v2, a sentence-transformer model producing a 384-dimensional vector for each passage, and the resulting vectors are normalised to unit length. The reference corpus was embedded with the same model at the time the index was built, so that reference and snapshot passages occupy a common vector space and the inner product between any two vectors equals their cosine similarity.

Retrieval is performed against a FAISS index holding the 6,916 reference vectors, using a range search rather than a nearest-neighbour search. For each snapshot passage the index returns every reference passage whose cosine similarity is 0.60 or above, so a passage may be paired with several reference passages or with none. A nearest-neighbour search would return a fixed number of results for every passage regardless of similarity, which at this scale would produce a pairing for each of the tens of millions of passages examined; the range search returns nothing where nothing is sufficiently similar (Douze et al., 2024).

The similarity threshold of 0.60 is used as a retrieval cutoff rather than as a criterion for deciding whether a pair constitutes a reproduction. Its purpose is to limit the number of candidate pairs passed to the next stage while retaining pairs that could satisfy the final matching criteria. The cutoff is intentionally conservative: a page may reproduce only a headline and opening sentences, while surrounding navigation or unrelated site content reduces the overall semantic similarity to the reference passage. Raising the cutoff could remove such genuine reproductions

before they reach the lexical verification stage, while a lower cutoff would substantially increase the number of candidates requiring further processing.

3.8 Stage 3: Containment

In the final stage, each pair returned by the retrieval stage is scored for verbatim overlap. Both passages are normalised by the procedure applied at the screening stage, and each is then decomposed into the set of all overlapping sequences of five characters it contains. The unit of comparison therefore changes between the two lexical stages: the screen matched sequences of eight words, whereas this stage matches sequences of five characters. Character sequences are preferred here because the small orthographic differences that commonly survive republication: a hyphen removed, a compound split, a space inserted, are narrower in their effect. A single such alteration destroys every word sequence spanning it, up to eight in the screening stage, which in a passage of thirty words is roughly a third of the sequences it contains; the same alteration destroys at most five character sequences out of the approximately two hundred that a passage of that length yields. The screen can absorb that loss because a page needs to share only one sequence to be retained, whereas this stage measures how much of a passage is reproduced and is therefore sensitive to how much of it survives intact.

Overlap is measured by containment where for two passages a and b, let $S(a)$ and $S(b)$ denote their respective sets of character 5-shingles. Containment is defined as:

$$C(a,b) = \frac{|S(a) \cap S(b)|}{\min\big(|S(a)|, |S(b)|\big)} \qquad (1)$$

where $|\cdot|$ denotes set cardinality. The measure ranges from 0, where the two passages share no five-character sequence, to 1, where every sequence of the shorter passage also occurs in the longer one.

The denominator is the size of the smaller set rather than of the union, and this distinction is essential given the nature of the target text. A page in the snapshot arrives as unsegmented plaintext, so a reproduced passage is typically accompanied by navigation, related-link lists, and footers belonging to the page rather than to the article. Dividing by the union, as the Jaccard coefficient does (Broder, 1997), would allow that surrounding material to dominate the denominator, and a passage reproduced word for word inside a long page would receive a low score.

No threshold is applied at this stage. Every pair returned by the retrieval stage is written together with its containment score, so that the threshold can be placed by reference to the resulting distribution rather than assumed in advance.

3.9 Threshold determination

The containment threshold was placed by reference to the distribution of scores the measure actually produces, rather than fixed in advance. A test run was carried out for this purpose: the full 2026 snapshot was screened and verified against a smaller reference corpus of 1,385 passages, and every resulting pair was retained with its containment score, yielding approximately 2.4 million scored pairs. The purpose of this run was to establish where the threshold should sit before the full corpus was run, not to produce findings. Figure 1 shows the distribution.

The distribution is unimodal: scores concentrate at low values, with a median of 0.14 and a ninety-ninth percentile of 0.36, and the density falls away sharply above 0.40. Above that point the distribution does not resolve into a second cluster but continues as a sparse, thinly populated continuum extending to 1.00. There is no empty band separating a population of matches from a population of non-matches, and consequently no automatic

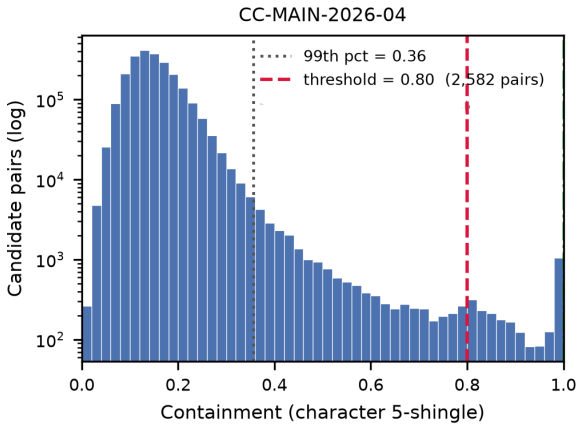


Figure 1: The distribution of containment scores in order to set the threshold

procedure that places the threshold at the trough between them.

This absence is itself a result rather than a failure of the measure. Republication is not a binary phenomenon: a third-party page may reproduce an entire passage, a headline and opening sentence, or a single paraphrased line, and the containment scores such pages produce form a continuum accordingly. A threshold therefore does not separate two natural populations but selects a level of confidence, and the question is where to place it rather than where to find it.

The threshold was set at 0.80, which lies 0.44 above the ninety-ninth percentile of the distribution and well clear of the region in which topically related but independently written text falls. A pair meeting it reproduces at least four fifths of the shorter passage's character sequences and is not plausibly the product of coincidental overlap. The value was chosen for the separation it provides from the noise, and was not adjusted to accommodate any particular pair.

The threshold was also validated by manual inspection rather than by its position in the distribution alone. A stratified random sample of 100 candidate pairs was drawn from both snapshots combined, twenty from each of five containment bands between 0.40 and 1.00, and each pair was labelled by hand according to whether it reproduced the reference passage. Pairs in which both texts reproduced the same underlying document, such as an official statement or a press release, were labelled as independent publication of a shared source rather than as reproduction of the publisher's text. Precision was 0.95 in each of the two bands at and above the threshold (19 of 20 pairs in each), 0.85 in 0.70–0.79 (17 of 20), 0.60 in 0.60–0.69 (12 of 20), and 0.15 in 0.40–0.59 (3 of 20). Pairs meeting the threshold are therefore reproductions in the large majority of cases. Below it, precision declines gradually in the adjacent band and steeply thereafter, which places 0.80 immediately above the point at which precision begins to fall.

Pooling the two bands at and above the threshold gives 38 reproductions in 40 labelled pairs, a precision of 0.95 with a 95% confidence interval of [0.83, 0.99]. The interval is computed with the Wilson score method, which remains reliable for small samples and for proportions close to one (Wilson, 1927). Its width reflects the size of the labelled sample, and it describes how precisely the error rate of the matching criteria is known rather than any variability in the occurrence counts themselves. The two errors among these pairs were not topically similar text but recurring boilerplate and a political statement quoted independently by both sources. This indicates that the residual error at this level arises from shared text of a kind other than reproduction rather than from coincidental overlap. Furthermore, the estimate applies to passage pairs, which are the unit that was labelled, and does not transfer directly to occurrences. An occurrence is formed by collapsing all pairs that link one article to one host (Section 3.10), so a single occurrence is often supported by several pairs. The proportion of correct occurrences may therefore differ from the proportion of correct pairs.

3.10 Analysis plan

The unit of analysis is an *occurrence*, defined as the presence of one reference article on one host. Multiple passages, pages, or URLs containing text from the same article on the same host are treated as a single occurrence; thus, if one article appears on twenty hosts, it contributes twenty occurrences, whereas two different articles on the same host contribute two occurrences. Hosts are determined from the host component of the URL, with only the removal of a leading `www.` prefix before comparison.

For the purpose of classifying occurrences, a host is treated as first-party when it is the publisher's registrable domain, `nature.com`, `theguardian.com`, or `nytimes.com`, or a subdomain of one of these domains. Every other host is classified as third-party. Classification and counting operate at different levels and are therefore independent: first-party status is determined in relation to the publisher's registrable domain, whereas occurrences are counted separately for each full host. A subdomain is consequently classified as first-party while remaining a distinct host for counting purposes, since it may in principle serve its own `robots.txt` file and therefore need not express the same reservation as its parent domain. Hosts associated with the same corporate group but using a different registrable domain are classified as third-party, because the reservation examined in this study is the one published on the publisher's own domain.

The same rule applies to regional editions and secondary sites: those served from a path or subdomain of the publisher's registrable domain are classified as first-party, and those served from a separate registrable domain are classified as third-party. Redirects are not resolved, since each page is attributed to the URL under which Common Crawl recorded its plaintext; text reached through a redirect is therefore attributed to the host that served it rather than to the URL that pointed to it.

An occurrence is counted only when a matched pair satisfies both matching criteria: a cosine similarity of at least 0.60 and a containment score of at least 0.80 (Sections 3.7 and 3.9). Pairs falling below either threshold are excluded, so the reported figures represent a conservative estimate of occurrences rather than an exhaustive count of all reproductions. Under these criteria the pipeline achieves high precision on the labelled pairs (Section 3.9), although it is not free of error in either direction. It may miss genuine reproductions, whether at the screening stage or among pairs that fall below one of the thresholds, and it may also retain a small number of false positives, such as the recurring boilerplate and the independently quoted statement identified in the manual validation.

The verification stage produces one row for each matched passage pair. Because snapshot windows advance in two-sentence steps and therefore overlap, and because an article may appear on multiple URLs within a domain, a single reproduction often generates several such pairs. To obtain the occurrence measure, these passage-level pairs are collapsed into the number of distinct combinations of publisher, reference-article identifier, and domain. All pairs associated with the same article on the same domain therefore count as one occurrence, irrespective of how many passages, windows, or URLs generated them. No equivalent deduplication is performed within the reference corpus: each source provides a unique identifier for every record, and the article identifiers were verified to be distinct within their respective sources.

Occurrences that satisfy both criteria are referred to throughout as *detected occurrences*. The term *verified occurrence* is used only for detected occurrences whose corresponding page–reference pair was additionally examined manually, as described

in Section 3.9.

The reference corpus is restricted to articles published on or before 31 December 2017. This restriction ensures that the same fixed set of reference articles is searched for in both snapshots, so any difference in results reflects variation in the contents of the two snapshots rather than changes in the reference set.

Occurrences on publishers' own domains and on third-party domains are both retained and reported separately. An occurrence on `nytimes.com`, for example, indicates that Common Crawl captured the publisher's own host during the relevant crawl period, whereas an occurrence on a third-party domain indicates that the article was reproduced elsewhere. Both types of occurrence provide evidence for H1. The distinction is nevertheless retained because it provides important context for interpreting the change between snapshots in H2.

Raw occurrence counts are not comparable across the three publishers, since they contribute 998, 220 and 2,404 reference articles respectively. Let A_p denote the set of reference articles contributed by publisher p, and $O_{p,s}$ the set of occurrences of those articles in snapshot s. Results are normalised as

$$\rho_{p,s} = \frac{|O_{p,s}|}{|A_p|} \tag{2}$$

where $|\cdot|$ denotes set cardinality: the number of occurrences per reference article contributed. A value of 1 represents an average of one occurrence per reference article; values above 1 indicate that articles were found on several domains each, and values below 1 that most articles were not found at all.

The two snapshots contain 3.10 and 2.30 billion pages, so raw counts are not directly comparable between them either. Normalising by snapshot size would treat crawl volume as the quantity against which republication should be measured, but crawl volume is a property of Common Crawl's own selection policy rather than of the publishers' content. The reference corpus, by contrast, is fixed: the same 3,622 articles are searched for in both snapshots, so occurrences per reference article are comparable across the two by construction. Raw occurrence counts are reported alongside the rate, to allow comparison using alternative normalisation choices.

Table 4: Primary parameter configuration. The same values are used for all results reported in Section 4.

Parameter	Value
Screening	
Sequence length	8 words
Retention	1 sequence
Windowing	
Window	4 sentences
Step, snapshot	2 sentences
Step, reference	4 sentences
Retrieval	
Cosine floor	0.60
Verification	
Shingle	5 characters
Containment	0.80

In order to address the hypotheses posed in Section 1, the same primary configuration of the pipeline, given in Table 4, is used for all three publishers in all results reported in Section 4. The sequence length and retention criterion were chosen for the reasons set out in Section 3.6, the windowing and the cosine floor in Section 3.7, and the shingle size and containment threshold in Sections 3.8 and 3.9.

Consequently, the reported counts characterise the complete searchable contents of the two selected corpora, subject to the recall and precision limitations of the matching pipeline. Because the analysis covers the full corpora rather than a sample, uncertainty arises from measurement error in the pipeline rather than from sampling variability; confidence intervals and inferential statistical tests are therefore not

reported. The comparison between snapshots is descriptive, quantifying the difference in detected occurrences between the two corpora without making inferences to a broader population.

For each publisher, $\rho_{p,s}$ is reported for both snapshots together with the difference between the two values. Comparisons are made separately for each publisher rather than on the pooled total, because the three publishers contribute passages of substantially different lengths and are therefore not equally detectable by the matching procedure. Since the comparison concerns reproduction on hosts the reservation does not reach, $O_{p,s}$ is restricted to occurrences on third-party domains wherever $\rho_{p,s}$ is used to compare the two snapshots.

H1 is supported when at least one detected occurrence on any domain is identified in either snapshot. A single qualifying occurrence is sufficient because H1 concerns the possibility that opted-out content is present, not its prevalence or rate. H2 is supported descriptively when $\rho_{p,2026}$ differs from $\rho_{p,2018}$ for at least one publisher. The direction and magnitude of each difference are reported without assigning it to a particular cause, because the present design cannot identify the causal mechanism responsible for a change between snapshots.

Neither hypothesis is treated as established solely because it holds under one parameter configuration. For H1, robustness is assessed by recording whether at least one third-party occurrence remains present under every configuration in the parameter grid reported in Section 3.11. For H2, robustness is assessed by recording whether the direction of the difference between snapshots remains unchanged across that grid, which is computed on third-party occurrences alone. The alternative configurations therefore function as a sensitivity analysis for the primary results, not as independent results on which separate conclusions are based.

3.11 Parameter sensitivity

A sample of 100 candidate pairs was labelled by the author where pairs were drawn from the full set of scored pairs (both snapshots combined) and stratified only by containment band. Thus, five bands (0.40–0.59, 0.60–0.69, 0.70–0.79, 0.80–0.89, 0.90–1.00) with twenty pairs were sampled at random within each band using a fixed random seed. The labelling sheet contained only the two passage texts and their source URLs; containment and cosine scores were withheld, and rows were shuffled so that band membership could not be inferred from order. Scores were rejoined only after labelling was complete. Each pair was classified as either a reproduction of the publisher's text or an independent publication of shared source material (for example, a press release, official statement, or quotation appearing in both passages). Where the initial judgement was uncertain, the full source pages were consulted manually on the publisher's website and in the Common Crawl plaintext to make a final determination based on the complete context.

The sensitivity grid is computed using third-party occurrences alone. Whether a publisher's own hosts appear in a snapshot is determined by the reservation state documented in Table 3, which differs between the two crawl periods. First-party counts therefore reflect the interaction between host availability and the crawl-time reservation state, rather than the placement of either matching threshold. Restricting the grid to third-party occurrences ensures that both snapshots contribute the same category of occurrence at every parameter setting, allowing the sensitivity of the republication comparison to be assessed independently of first-party host coverage.

The cosine threshold test serves a purpose beyond confirming robustness. Rais-

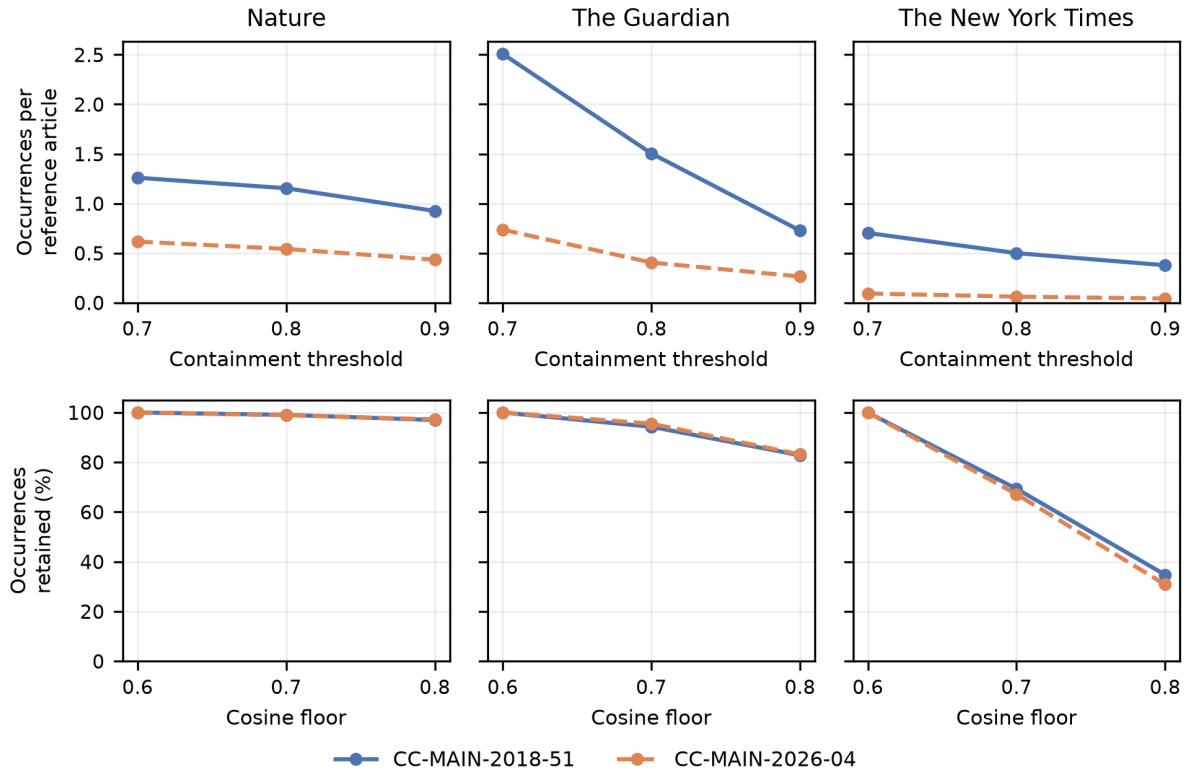


Figure 2: Sensitivity of occurrence counts on domains other than the publisher's own. Top row: occurrences per reference article at containment thresholds of 0.70, 0.80 and 0.90, with the cosine floor held at 0.60. Bottom row: occurrences retained at containment 0.80 as the cosine floor is raised, as a percentage of the count at 0.60.

ing the cosine floor from 0.60 to 0.80 eliminates 3% of Nature's occurrences and 17% of The Guardian's, but removes 65% of The New York Times's occurrences in the 2018 snapshot and 69% in the 2026 snapshot. The excluded pairs are not weak matches: they satisfy the containment threshold and reproduce the reference passage verbatim. This asymmetry arises from differences in passage length. A New York Times reference record typically consists of a headline and lead paragraph of approximately thirty words; when such a short passage appears within a page that also contains navigation elements, related-link lists, and unrelated articles, the similarity between the full page window and the reference passage is diluted, even when the reproduction is exact. The low floor adopted in Section 3.7 was chosen precisely to accommodate such cases: the test quantifies what would have been discarded under a higher threshold.

Figure 2 reports the grid, one publisher per column. Nature is the least affected by either parameter: its rate falls by a fifth across the containment range and its counts by three percent across the cosine range. The Guardian is the most sensitive to the containment threshold, its rate falling from 2.51 to 0.73 occurrences per reference article in the 2018 snapshot, and moderately sensitive to the cosine floor. The New York Times is the reverse case, falling by less than half across the containment range but losing roughly two thirds of its occurrences when the cosine floor is raised.

3.12 Reproducibility and compute environment

The pipeline is implemented in Python and organised as a sequence of command-line scripts, each writing its output to disk before the next is run, so that any

stage can be repeated without recomputing those preceding it. Lexical screening uses the pyahocorasick implementation of the Aho–Corasick automaton; embedding uses the sentence-transformers implementation of all-MiniLM-L6-v2; retrieval uses FAISS with an exact inner-product index. All parameters governing the analysis are fixed rather than inferred at run time; the values are listed in Section 3.10.

The lexical screening and semantic retrieval have different computational profiles and were run on different machines. Screening reads every archive file in a snapshot from object storage and is bound by input and processing throughput rather than by memory; it was run on a 64-core cloud instance in the same region as the archive, taking approximately fifteen hours for the 2026 snapshot and nineteen hours for the 2018 one. Verification embeds the retained pages and is bound by the throughput of a single accelerator; it was run locally on a machine with GPU acceleration. Because the screen retains the extracted text of each surviving page, verification requires no access to the archive and can be repeated independently of it.

The reference corpus is reproducible from its sources. The collection scripts record the queries, date bounds, and per-publisher targets used, and the three sources are public interfaces rather than private datasets, so the corpus can be regenerated from the same parameters. The snapshots are permanent public archives identified by name, and both remain available in the form analysed.

4 Results

4.1 Nature

Table 5 presents the detected occurrences of Nature reference articles across the two crawl snapshots. In both datasets, Nature articles were detected on domains other than the publisher's official site. The December 2018 snapshot contained 1,152 such occurrences, whereas the January 2026 snapshot contained 541. When normalised by the number of reference articles, the rate declines from 1.154 occurrences per reference article in 2018 to 0.542 in 2026, corresponding to a decrease of 0.612.

Furthermore, 103 occurrences appear on the publisher's own domain in the December 2018 snapshot, all of them on `nature.com`, which is the only first-party host carrying Nature text in either snapshot. Because a single domain is involved, this figure is also the number of distinct reference articles captured there, corresponding to 0.103 occurrences per reference article contributed. No occurrence on a domain owned by the publisher appears in the January 2026 snapshot.

The occurrences are distributed across a large number of separate hosts. In the December 2018 snapshot, 535 of the 998 Nature reference articles were found at least once on a third-party domain, and those articles appear across 545 distinct domains; in the January 2026 snapshot the corresponding figures are 322 articles across 332 domains. In both snapshots the median number of domains carrying a given found article is one, while the maximum is 11 in the earlier snapshot and 7 in the later one.

Two verified occurrences, one from each snapshot, are presented below to illustrate the form of the matched text. These examples are illustrative rather than representative: each was selected from the manual inspection sample, and no inferential weight is attached to the specific cases chosen. In the December 2018 snapshot, an abstract concerning colour change in panther chameleons was matched on `previouslife.lanevol.org`, a research group's project page, with a containment score of 0.974 and a cosine

Table 5: Detected occurrences of Nature reference articles in the two snapshots. One occurrence is one reference article present on one domain. Occurrences on the publisher's own domains are reported separately as crawl context.

	CC-MAIN-2018-51	CC-MAIN-2026-04
Third-party domains		
Occurrences	1,152	541
ρ[a]	1.154	0.542
Articles found, n	535	322
Articles found, %	53.6	32.3
Domains per article, median	1	1
Domains per article, max	11	7
Domains	545	332
Publisher's own domains (crawl context)		
Occurrences	103	0

[a] ρ is the number of third-party occurrences divided by the number of reference articles the publisher contributes, as defined in Equation 2. The denominator here is the 998 Nature reference articles, and it is the same in both columns, so the two values are directly comparable. First-party occurrences are excluded from ρ because the comparison between snapshots is made on third-party occurrences alone (Section 3.10).
Domains per article covers articles found at least once on a third-party domain.

similarity of 0.988.[1] The two texts differ only in minor editorial details: the web page includes the species' binomial name, whereas the reference omits an introductory adverb.

In the January 2026 snapshot, an abstract on biodiversity and ecosystem multifunctionality was matched on `scholarworks.wm.edu`, the institutional repository of the College of William & Mary, with a containment score of 1.000 and a cosine similarity of 1.000.[2] The repository record reproduces the abstract in full as part of its bibliographic entry for the article. Neither domain belongs to the publisher whose text was matched.

4.2 The Guardian

Table 6 reports the detected occurrences of Guardian reference articles in the two snapshots. Guardian articles were found on domains other than the publisher's own in both, with 331 such occurrences in December 2018 and 89 in January 2026. The normalised rate falls from 1.505 to 0.405 occurrences per reference article between the two crawls, a difference of −1.100.

Moreover, 56 occurrences appear on the publisher's own domains in the December 2018 snapshot, of which 55 are on `theguardian.com` and one on `amp.theguardian.com`, each counted as a separate host (Section 3.10). Together they correspond to 0.255 occurrences per reference article contributed. No occurrence on a domain owned by the publisher appears in the January 2026 snapshot.

The occurrences are concentrated on a smaller set of articles than for Nature. In the December 2018 snapshot, 81 of the 220 Guardian reference articles were found at least once on a third-party domain, and those articles appear across 302 distinct domains; in the January 2026 snapshot the corresponding figures are 32 articles

[1] `http://previouslife.lanevol.org/LANE/chameleon_colour_change.html`; reference record `https://openalex.org/W1983416666`

[2] `https://scholarworks.wm.edu/entities/publication/1f50efb3-0e6b-4a29-b049-dadc529538ef`; reference record `https://openalex.org/W1980525736`

Table 6: Detected occurrences of The Guardian reference articles in the two snapshots. One occurrence is one reference article present on one domain. Occurrences on the publisher's own domains are reported separately as crawl context.

	CC-MAIN-2018-51	CC-MAIN-2026-04
Third-party domains		
Occurrences	331	89
ρ[a]	1.505	0.405
Articles found, n	81	32
Articles found, %	36.8	14.5
Domains per article, median	1	1
Domains per article, max	121	40
Domains	302	82
Publisher's own domains (crawl context)		
Occurrences	56	0

[a] ρ is the number of third-party occurrences divided by the number of reference articles the publisher contributes, as defined in Equation 2. The denominator here is the 220 Guardian reference articles, and it is the same in both columns, so the two values are directly comparable.
Domains per article covers articles found at least once on a third-party domain.

across 82 domains. The median number of domains carrying a given found article is one in both snapshots, while the maximum is 121 in the earlier snapshot and 40 in the later one.

In the December 2018 snapshot, a four-sentence window from a report on the arrest of a German intelligence officer was matched on `israelagainstterror.blogspot.com` with a containment score of 1.000 and a cosine similarity of 1.000.[3] The window appears on the page without alteration, on a dated archive page of a personal blog.

In the January 2026 snapshot, a four-sentence window from a report on a military raid in Yemen was matched on `tenthltr2u.com` with a containment score of 1.000 and a cosine similarity of 1.000.[4] The window is reproduced without alteration and without attribution to the publisher. Neither domain belongs to the publisher whose text was matched.

4.3 New York Times

Table 7 reports the detected occurrences of New York Times reference articles in the two snapshots. New York Times articles were found on domains other than the publisher's own in both, with 1,203 such occurrences in December 2018 and 146 in January 2026. The normalised rate falls from 0.500 to 0.061 occurrences per reference article between the two crawls, a difference of -0.439.

Furthermore, 444 occurrences appear on the publisher's own domains in the December 2018 snapshot, distributed over four hosts: 440 on `nytimes.com`, two on `dealbook.nytimes.com`, one on `archive.nytimes.com` and one on

[3] http://israelagainstterror.blogspot.com/2016_11_27_archive.html; reference article https://www.theguardian.com/world/2016/nov/30/german-intelligence-agent-held-over-leaks-to-islamic-extremists

[4] https://www.tenthltr2u.com/post/questions-mount-over-botched-yemen-raid-approved-by-trump; reference article https://www.theguardian.com/world/2017/feb/02/trump-approved-yemen-raid-five-days-after-inauguration

Table 7: Detected occurrences of The New York Times reference articles in the two snapshots. One occurrence is one reference article present on one domain. Occurrences on the publisher's own domains are reported separately as crawl context.

	CC-MAIN-2018-51	CC-MAIN-2026-04
Third-party domains		
Occurrences	1,203	146
ρ[a]	0.500	0.061
Articles found, n	610	118
Articles found, %	25.4	4.9
Domains per article, median	1	1
Domains per article, max	30	4
Domains	740	118
Publisher's own domains (crawl context)		
Occurrences	444	0

[a] ρ is the number of third-party occurrences divided by the number of reference articles the publisher contributes, as defined in Equation 2. The denominator here is the 2,404 New York Times reference articles, and it is the same in both columns, so the two values are directly comparable. Domains per article covers articles found at least once on a third-party domain.

`lens.blogs.nytimes.com`. Because each host is counted separately (Section 3.10) and two reference articles were found on more than one of them, the 444 occurrences correspond to 442 distinct articles. Together they amount to 0.185 occurrences per reference article contributed. No occurrence on a domain owned by the publisher appears in the January 2026 snapshot.

The occurrences are distributed across a large number of separate hosts. In the December 2018 snapshot, 610 of the 2,404 New York Times reference articles were found at least once on a third-party domain, and those articles appear across 740 distinct domains; in the January 2026 snapshot the corresponding figures are 118 articles across 118 domains. The median number of domains carrying a given found article is one in both snapshots, while the maximum is 30 in the earlier snapshot and 4 in the later one.

In the December 2018 snapshot, the record for *Uber Hit With New Blow in London as Panel Says Drivers Aren't Self-Employed* was matched on `bostonglobe.com` with a containment score of 0.801 and a cosine similarity of 0.843.[5] The page carries the headline, the byline *By Prashant S. Rao*, the credit line *New York Times*, and the opening paragraphs of the article. The matched text spans the headline and the passage immediately following it.

In the January 2026 snapshot, the record for *Careers for Women in Technology Companies Are a Global Challenge* was matched on `dokasi.com` with a containment score of 1.000 and a cosine similarity of 0.674.[6] The headline and the sentence following it are reproduced verbatim in a post on a weblog, surrounded by site navigation, a category archive listing, and unrelated post titles. Neither domain

[5] `https://www.bostonglobe.com/business/2017/11/10/uber-hit-with-new-blow-london-panel-says-drivers-aren-self-employed/uyh5BekxIdf3123wPqpVXO/amp.html`; reference article `https://www.nytimes.com/2017/11/10/business/uk-uber-london.html`

[6] `https://www.dokasi.com/news/women-today-careers-for-women-in-technology-companies-are-a-global-challenge/`; reference article `https://www.nytimes.com/2017/10/10/business/women-careers-technology-companies.html`

belongs to the publisher whose text was matched.

4.4 Hypothesis evaluation

H1 is supported, since the criterion set out in Section 3.10, at least one detected occurrence on any domain in either snapshot, is satisfied. Detected occurrences on third-party domains were present for all three publishers in both snapshots: 1,152 in CC-MAIN-2018-51 and 541 in CC-MAIN-2026-04 for Nature, 331 and 89 for The Guardian, and 1,203 and 146 for The New York Times (Tables 5, 6, 7). Occurrences on the publishers' own domains, by contrast, were present in CC-MAIN-2018-51 only. As detected rather than manually verified occurrences, these counts may include a small number of false positives (Section 3.9).

H2 is descriptively supported for the selected reference articles and the two snapshots examined, since the criterion set out in Section 3.10, a difference in $\rho_{p,s}$ between the two snapshots for at least one publisher, is met for all three publishers. For Nature, $\rho_{p,s}$ declines from 1.154 to 0.542, a decrease of 0.612 (53.0%); for The Guardian, from 1.505 to 0.405, a decrease of 1.100 (73.1%); and for The New York Times, from 0.500 to 0.061, a decrease of 0.439 (87.9%). These rates are observed detection rates rather than direct measures of how frequently each publisher's content is republished, and they are not directly comparable across publishers, because the reference passages differ in length and therefore in how readily a reproduction can be detected (Section 3.3). At the same time, first-party occurrences fall from 603 to zero (a 100% reduction), while third-party occurrences decline from 2,686 to 776 (a 71.1% reduction).

5 Discussion

5.1 Implications of Hypothesis 1

The results establish that content later covered by the publishers' machine-readable reservations was detected in both Common Crawl snapshots, including on domains other than those controlled by the publishers. The implication of this finding is not that every item of opted-out content enters the archive, nor that the archive necessarily supplies that content to a particular AI model. Rather, it shows that a machine-readable signal on the publisher's own host does not, by itself, keep copies of its articles out of a corpus assembled from the web.

This result is consistent with the host-level operation of `robots.txt` where the file is consulted when a crawler requests a page from the host on which it is located, and it does not travel with the article when the article is copied, quoted, republished, or deposited elsewhere. The machine-readable signal published on the original host is therefore not automatically available to a crawler collecting the copy from another host. Thus, a publisher can block Common Crawl from its own domain while the same text is collected from a host that publishes no equivalent signal. The 3,462 third-party occurrences identified across the two snapshots show that this is not limited to a single anomalous page: for the selected reference corpus, copies consistent with this potential collecting route were repeatedly across all three publishers.

The finding should nevertheless be interpreted within the scope of the mechanism examined. It does not establish that any particular training dataset or AI model contains the identified passages. Presence in Common Crawl makes the content available for possible inclusion in derived corpora, but whether it is included depends on the processing applied by each provider. The derived corpora reviewed in Section 1 document filters for language, quality, and duplication, but no procedure for identifying whether text hosted on a third-party domain originated with a publisher that has reserved its

rights (Raffel et al., 2020; Abadji et al., 2022; Penedo et al., 2023, 2024). The retroactive check described in Section 2.2 likewise remains host-level (Fan et al., 2025), because the Robots Exclusion Protocol applies to the host serving a resource and does not establish the provenance of text reproduced on another host (Koster et al., 2022).

The results therefore identify an information gap for any actor seeking to apply a reservation to copies of content after republication. A check of the host from which a document was collected cannot determine whether that document reproduces text from a publisher that has expressed a reservation elsewhere. This limitation is structural, arising from the fact that the reservation is expressed at the level of individual hosts while the archive assembles copies drawn from many different hosts. The study does not determine which actor should address this gap, but it shows that host-level checks alone do not make third-party reproductions visible.

The gap carries implications for policy where the Directive leaves the choice of means open, but the protocol most commonly used to express a reservation, and the one named explicitly in the Code of Practice, operates at the level of individual hosts, while the AI Act delegates to providers the task of identifying and complying with reservations (European Parliament and Council of the European Union, 2019, 2024; European Commission, AI Office, 2025a). Whether additional obligations, guidance, or technical measures should address this gap is a policy choice, but the study identifies a condition that any such measure would need to overcome.

5.2 Implications of Hypothesis 2

The results show that the number of occurrences differs between the two snapshots for every publisher, and that the direction of this difference remains unchanged across the tested parameter configurations. The implication of H2 is therefore that the selected reference corpus has a different observable presence in the two examined archives. This is, in the first instance, a descriptive finding about the contents of the snapshots. The comparison does not permit the overall difference between snapshots to be attributed conclusively to the Directive, the publishers' reservations, changes in Common Crawl's crawling policy, or any other individual factor. Nevertheless, the documented reservation states and the distribution of first-party occurrences provide contextual evidence relevant to the interpretation of one component of the observed change.

The difference is best understood as the combination of two distinct patterns. The first is the disappearance of occurrences on the publishers' own domains; the second is the decline, but not elimination, of occurrences on third-party domains. These movements carry different implications because a `robots.txt` directive is evaluated only on the host that publishes it. Treating the total change as a single phenomenon would obscure the fact that the signal can prevent collection from the publisher's own host while remaining unavailable to a crawler collecting copies from other hosts.

The first pattern is the disappearance of first-party occurrences from the later snapshot. In the earlier snapshot, occurrences were identified on the publishers' own domains; no such occurrences were identified in the later snapshot. This pattern is consistent with the documented reservation state of the publishers' hosts during the later crawl period. Where the relevant `robots.txt` files disallowed Common Crawl's crawler, requests to those hosts would not have been permitted under the crawler's stated rules. The comparison cannot establish that the reservations caused the disappearance of first-party occurrences, since other changes between the crawl periods cannot be excluded. It

nevertheless indicates that the observed first-party pattern is compatible with the host-specific operation of the `robots.txt` signal.

The second movement is the decline in occurrences on third-party domains between the two snapshots but did not disappear for any of the three publishers. This persistence has a different implication from the disappearance of first-party occurrences. The `robots.txt` signal published on the publishers' own hosts is not automatically available to a crawler collecting the same text from a third-party website, aggregator, repository, or other host. The later snapshot consequently shows that a signal on the original host does not prevent copies of an article held elsewhere from being collected.

The decline in third-party occurrences is not attributed to the publishers' reservations. Its direction is not an artefact of the particular cosine and containment thresholds tested, but the design cannot exclude other explanations for it, which are set out in Section 5.4.

The implication of H2 is therefore not that the study proves the Directive caused the observed change. Rather, the comparison shows that the archive's observable exposure of the selected reference corpus changed across the two crawl periods, while the mechanisms associated with first-party and third-party hosts behaved differently. The first-party change is strongly consistent with the operation of the publishers' `robots.txt` signals; the third-party change is an observed overall pattern for which this study does not identify a single cause. Separating these patterns prevents the complete disappearance of first-party occurrences from being mistaken for the complete removal of the underlying content from the archive.

5.3 Policy Implications for the Development of Large Language Models

The results discussed in Sections 5.1 and 5.2 indicate a potential collecting route by which opted-out content can enter the archive. The route is inferred from the presence of copies on third-party hosts rather than observed at the moment of collection. This section considers what follows from that potential route for the framework that currently governs the composition of AI training sets. The argument concerns Common Crawl itself, not the training datasets derived from it or the developers that use them. Presence in Common Crawl makes the content available for possible inclusion in derived corpora; whether it is included depends on how each corpus is constructed, which this study does not examine.

Alongside the policy obligation described in Section 1, Article 53(1)(d) of the AI Act requires every provider of a general-purpose AI model to draw up and make publicly available a summary of the content used to train it, following a template issued by the AI Office (European Parliament and Council of the European Union, 2024; European Commission, AI Office, 2025b). For data scraped from the web, the template asks for the top 10% of domain names determined by the size of content scraped, or, for small and medium-sized enterprises, the top 5% or 1,000 domains, whichever is lower, and Recital 107 describes a summary that is generally comprehensive in scope rather than technically detailed (European Commission, AI Office, 2025b; European Parliament and Council of the European Union, 2024). Because the template requires only the most prominent scraped domains rather than an exhaustive host list, small third-party hosts such as tenthltr2u.com, scholarworks.wm.edu or dokasi.com are unlikely to be individually visible in the summary. The occurrences recorded in Section 4 are therefore likely to fall below the res-

olution of the instrument through which the framework makes training content visible.

Recital 107 as well as 108 are strengthening the argument while also delimiting this inference. Recital 107 presents the summary as a means for parties with legitimate interests, including rightholders, to exercise and enforce their rights, and Recital 108 specifies that the AI Office monitors whether providers have fulfilled their obligations without verifying or assessing the training data work by work for copyright compliance (European Parliament and Council of the European Union, 2024). Identifying individual copies is therefore left largely to rightholders, which increases the significance of copies that the summary is unlikely to show. At the same time, Recital 108 states that the AI Act does not affect the enforcement of copyright rules under Union law, and the summary was not designed as a compliance audit. Its limited resolution is therefore a limitation of the transparency instrument, not evidence that any provider has failed to comply with a reservation.

Consequently, the 2018 snapshot illustrates a related temporal property of the archive. The 603 occurrences recorded on the publishers' own domains in Section 4 were collected when none of the three publishers disallowed the Common Crawl crawler, as Table 3 records. Those hosts now disallow the crawler across the whole domain, yet the earlier snapshot still contains the pages collected before. The archived snapshot records the `robots.txt` state at collection time and is not automatically updated when a publisher later changes its reservation, consistent with the observation of Fan et al. (2025) that Common Crawl's opt-outs are enforced only at the time of crawling. A crawler following the current `robots.txt` files would not collect these pages today, whereas the earlier snapshot keeps them available for possible inclusion in corpora derived from it.

5.4 Limitations

The findings are constrained by the study design, beginning with the composition of the reference corpus. The articles in it were ordered by each API's own relevance or citation ranking rather than drawn at random, so the counts reported in Section 4 establish the presence of 3,622 specific articles. Consequently, they cannot be scaled into an estimate of how much reserved material either snapshot contains, since such an estimate would require a corpus assembled to represent the entirety of the opted-out content on the internet.

A second limitation concerns the decision rule: some shared-source cases survive both matching criteria. Where a reference passage consists largely of quoted statements (for example from the speech of a president), a page reproducing the same underlying document satisfies the containment threshold and the cosine floor without reproducing the publisher's own text, and the manual inspection described in Section 3.9 identified such pairs above the threshold. Furthermore, this residual error arises from the decision rule rather than from the data itself, since neither criterion distinguishes text a publisher wrote from text it quoted from a source available to every outlet, and it concentrates in The Guardian, whose four-sentence windows often cover passages in which the publisher is relaying statements made elsewhere rather than writing its own account. Section 5.5 sets out a possible fix that would separate the two cases. H1 does not depend on these pairs, as it holds for all three publishers in both snapshots. H2, however, is computed on third-party occurrences, which may include such cases. Its overall direction appears robust, as it is unchanged across the parameter grid in Section 3.11, while the precise counts and effect sizes

may be affected.

The precision estimate is also limited by its unit due to the fact that the estimate of 0.95 (95% CI 0.83–0.99) applies to sampled passage pairs, the unit that was labelled, whereas the unit reported in Section 4 is an aggregated article–host occurrence, which may be supported by several pairs (Section 3.10). Occurrence-level precision was not independently estimated, so the proportion of reported occurrences that are genuine reproductions may differ from the pair-level estimate in either direction.

Common Crawl's published plaintext is produced by a mechanical extraction that does not separate an article from the page carrying it, so navigation menus, related-link lists and footers appear in the text alongside the content, as described in Section 3.4. The containment measure was chosen to tolerate this, since dividing by the smaller shingle set leaves a verbatim passage at a high score however much surrounding material accompanies it, whereas the cosine similarity between a page window and a reference passage falls as that material grows. Genuine reproductions are therefore scored lower on one of the two criteria than a dedicated extractor would have scored them, and the effect is strongest for short reference passages, where the reproduced text occupies the smallest share of the window. A dedicated extractor would have removed the boilerplate at the cost of running a second extraction over every page in the snapshot, as discussed in Section 3.4.

Recall was not estimated, so the study cannot determine how many genuine reproductions were lost. Losses can occur at four points: during the word 8-gram screening, where a copy altered so that no unbroken run of eight words survives is discarded; below the cosine floor, where a short reproduction is diluted by surrounding page content; below the containment threshold, where a partial reproduction shares too few character sequences; and through the plaintext extraction, which lowers the scores of genuine reproductions as described above. The reported counts are therefore detected occurrences subject to an unknown miss rate, not complete counts of the reproductions present in either snapshot.

The comparison between the two snapshots is limited in what it can explain. The study compares two complete snapshots but does not control for factors that change between crawls, so the observed temporal difference may reflect changing web content, the survival or disappearance of domains, Common Crawl's crawl priorities, differences in extraction, or changes in page accessibility. The sensitivity analysis shows that the direction of the difference is stable across the tested matching parameters, but does not establish evidence of causation. The decline in third-party occurrences is therefore reported as an observed pattern without causal attribution.

Finally, the rates cannot be compared between publishers as measures of how often each publisher's content is republished. The reference passages differ greatly in length and source, ranging from full Guardian body text to New York Times records of a headline and abstract with a median of 32 words and Nature abstracts reconstructed from a bibliographic index (Sections 3.2 and 3.3), and a longer passage offers more opportunities for a reproduction to be detected. Differences in $\rho_{p,s}$ between publishers therefore reflect detectability as well as republication, and cannot be interpreted as direct differences in actual republication prevalence.

5.5 Future work

Several limitations identified in this study provide clear directions for future research. The shared-source residual is likely the most straightforward to address. This residual occurs when a reference passage

consists largely of material quoted by the publisher rather than original publisher-authored text. Because both the semantic similarity and containment measures capture textual overlap, they cannot distinguish republication of a publisher's prose from independent reproduction of the same underlying source, such as an official statement or speech. A potential solution would be to remove quoted material before articles are divided into passages and compared with Common Crawl pages. This would reduce the number of candidate pairs by limiting the reference corpus to publisher-authored prose, while pages reproducing only the underlying quoted material would no longer produce a match. Such preprocessing would, however, shorten some passages. Passages that subsequently fall below the fifteen-word threshold would need to be excluded, reducing recall for material that is already less suitable for identifying republication of the publisher's own text.

The validation of the pipeline could also be extended in three ways. First, a sample of occurrences rather than passage pairs could be labelled, so that precision is estimated for the unit the results report. Second, recall could be estimated by inserting known copies of reference passages, both exact and lightly altered, into a sample of WET files and recording how many each stage retains, together with a labelled sample of pairs below the thresholds. Third, re-extracting a sample of matched pages from the original WARC records with a dedicated extractor such as Trafilatura (Barbaresi, 2021) would show how strongly the boilerplate in Common Crawl's plaintext lowers the scores of genuine reproductions.

Another direction for future work is to adapt the matching thresholds to the type of content. A single cosine floor and containment threshold were applied to all three publishers, although the sensitivity analysis shows that news leads, full articles and scientific abstracts respond differently to each parameter (Section 3.11), and the three are republished in different ways, from syndicated copies of news reports to repository records of abstracts. Thresholds could therefore be set separately for each publisher or content type. Alternatively, the two fixed thresholds could be replaced by a classifier, such as a logistic regression that combines cosine similarity, containment and passage length into a single probability of reproduction, trained on manually labelled pairs. Either approach would require a larger labelled sample per content type than the 100 pairs used here.

The scope of the publisher sample also warrants expansion. The three publishers examined in this study represent news, general reporting, and academic research, and the republication observed may therefore reflect characteristics of these particular publishing sectors rather than reserved content more broadly. Literary publishing would provide an informative additional case, as it may involve different forms of republication, distribution, and rights management. Constructing a reference corpus for this sector would nevertheless be more difficult, since literary texts are generally not available through public APIs in the same way as the sources used in this study. Access would likely need to be granted directly by rightholders. Extending the reference corpus to literary publishing and other content sectors would make it possible to assess which observed patterns are general features of third-party republication and which are specific to particular types of publishing.

A further extension concerns the temporal resolution of the comparison. Two crawls establish that the snapshots differ, but they leave the intervening period unobserved, so the study cannot distinguish a gradual decline from one that occurred over a short interval within it. Screening additional snapshots between the two examined here would resolve that question

and describe the trajectory of reserved content in the archive rather than only its endpoints. The pipeline requires no modification for this, as the reference corpus is fixed and the same articles can be searched for in any snapshot, so the constraint is the screening cost of each additional crawl rather than the method.

6 Conclusion

Overall, the selected articles were identified in both Common Crawl snapshots on domains not controlled by their publishers. The 2018 snapshot contained 2,686 occurrences of articles later subject to the publishers' reservations, and the 2026 snapshot contained 776 occurrences of articles subject to reservations at crawl time, with occurrences associated with all three publishers in both crawls. The snapshots also differed with respect to the publishers' own domains: these domains accounted for 603 occurrences in the 2018 snapshot and none in the 2026 snapshot, which was crawled after all three publishers had disallowed `CCBot` in their `robots.txt` files.

The study therefore provides affirmative answers to both parts of the research question. A publisher's content can appear in Common Crawl despite a machine-readable reservation of text and data mining rights, and the observable presence of the selected articles differs between the two snapshots: occurrences on third-party domains per reference article were lower in the 2026 snapshot for all three publishers, and occurrences on the publishers' own domains, present in 2018, were absent in 2026. The disappearance of all occurrences on the publishers' own domains after those domains began disallowing `CCBot` is consistent with the `robots.txt` signal operating on the host that publishes it. The empirically supported conclusion, however, is that the original host's machine-readable signal was not visible from the third-party hosts on which the same articles were found. The findings consequently indicate a potential collecting route through which reserved content can enter Common Crawl, rather than establishing how frequently this occurs across publishers or the wider web. In addition, the observed decline in occurrences on third-party domains is reported without attributing it to a specific cause.

These findings relate directly to the broader context: the current framework relies on machine-readable reservations, which are most commonly expressed through `robots.txt`, a signal that operates at the level of individual web hosts. The evidence strongly supports a technical visibility gap between such host-level signals and content-level identification across hosts. A `robots.txt` signal can stop a compliant crawler from collecting pages from the host that publishes it, but it is not visible to a crawler collecting a copy from another host, which follows that host's own `robots.txt` file. As a result, such copies are not identified through the signal itself, and finding them requires content-level detection of the kind applied in this study. The results therefore indicate that the effectiveness of a host-level reservation cannot be evaluated solely by examining whether the reserving publisher's own domain remains in the crawl. For providers of derived training datasets, this suggests that host-level signals alone may not identify reserved content reproduced on third-party hosts, and that content-level checks could complement them.

7 Declaration on the use of Generative AI

During the preparation of this thesis, I used the following generative AI tool: Claude (Anthropic). The tool has been used for the following purposes: rephrasing, correcting table formatting and column layouts in LaTeX, inserting and adjusting figures and diagrams in Overleaf, and, throughout the semester, using Claude as a formative reviewer to assess

my work against the assessment rubric and identify gaps in my evidence portfolio for the intended learning outcomes (ILOs).

I did not use generative AI to generate arguments, results, or conclusions presented as my own; to fabricate, alter, or invent data; or to produce the core analysis of this thesis.

All research design, implementation, analysis, and conclusions are my own. I have critically reviewed and edited all content produced or supported by generative AI tools. I take full responsibility for the content and academic integrity of this thesis, including any sections drafted, refined, or supported with AI assistance.

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A Archived robots.txt captures

- https://web.archive.org/web/20181231004259/https://www.nature.com/robots.txt
- https://web.archive.org/web/20181231003146/https://www.theguardian.com/robots.txt
- https://web.archive.org/web/20181231004154/https://www.nytimes.com/robots.txt
- https://web.archive.org/web/20260101000749/https://www.nature.com/robots.txt
- https://web.archive.org/web/20260101011156/https://www.nytimes.com/robots.txt
- https://web.archive.org/web/20260101012243/https://www.theguardian.com/robots.txt

B Reference corpus acquisition

All three sources were collected with `01_pull_reference_corpus.py` under the run tag `full`, writing to `reference_corpus/<publisher>/full/`. The publication window was 2010-01-01 to 2017-12-31 in every case, and the same cutoff was applied a second time at index construction (`-date-cutoff 2018-01-01`), so a record published after 2017-12-31 is discarded whether or not the API honoured the date filter. Every collected record carries a `collected_at` UTC timestamp. Retrieval took place on April 2026.

B.1 The Guardian

Endpoint	`content.guardianapis.com/search`
Authentication	API key, Open Platform developer tier
Target	220 articles
Sections	`world`, `uk-news`, `us-news`, `politics`, `environment`, `science`, `technology`, `business`, `global-development`
Ranking	`order-by=relevance`
Pagination	`page-size=50`, incremented until the target or the reported final page
Fields	`headline`, `trailText`, `bodyText`, `wordcount`, `firstPublicationDate`; keyword tags
Rate limit	12 requests per second
Retained	220 articles, 2,040 passages

B.2 The New York Times

Endpoint	`api.nytimes.com/svc/search/v2/articlesearch.json`
Authentication	API key, developer tier
Target	2,500 records
Queries	`economy`, `climate change`, `technology`, `election`, `immigration`, `healthcare`, `education`, `foreign policy`, `supreme court`, `wall street`
Ranking	`sort=relevance`
Pagination	pages 0–99 per query, ten records per page
Fields	`headline`, `abstract`, `snippet`, `lead_paragraph`, `web_url`, `pub_date`, `section_name`, `word_count`, `source`, `_id`
Rate limit	one request per twelve seconds
Retained	2,404 records, 2,449 passages

The `lead_paragraph` field was requested but returned empty for every record in the 2010–2017 window; the field was withdrawn from the Article Search API in April 2025 and the withdrawal extends to the archive. Records therefore consist of a headline and a one- or two-sentence abstract, with a median passage length of 32 words. This is the source of the length asymmetry described in Section 3.2, and it was a property of the interface at retrieval time rather than a collection choice.

B.3 Nature

Nature abstracts were obtained from OpenAlex rather than from nature.com.

Endpoint	`api.openalex.org/works`
Authentication	none; `mailto` parameter for the polite pool
Target	1,000 records, divided equally across three journals ($\lfloor 1000/3 \rfloor = 333$ each)
Journals	Nature (`S137773608`), Nature Communications (`S64187185`), Scientific Reports (`S196734849`)
Filter	`primary_location.source.id`, `publication_year:2010-2017`, `has_abstract:true`, `type:article`
Ranking	`cited_by_count:desc`
Pagination	cursor, `per_page=100`
Fields	`id`, `doi`, `title`, `authorships`, `publication_date`, `primary_location`, `abstract_inverted_index`, `cited_by_count`
Retained	998 articles, 2,427 passages

OpenAlex stores abstracts as an inverted index mapping each token to its positions; running text was reconstructed by sorting tokens on position at collection time.

Two selection decisions constrain this source. A five-journal set was attempted first and abandoned: two of the source identifiers returned 404, and two of the remaining journals (Nature Machine Intelligence, launched 2019, and Nature Energy, launched 2016) hold no usable pre-2018 back-catalogue. A survey of the portfolio found three journals with more than two hundred abstracted pre-2018 articles, which are the three listed above. Second, the quota was split equally rather than in proportion to volume;

the pre-2018 abstracted counts at selection time were 62,602 for Scientific Reports, 12,932 for Nature Communications and 758 for Nature, so a proportional split would have drawn roughly four fifths of the corpus from a single journal.

B.4 Exclusion rules

Three filters were applied, at two stages.

1. **At collection.** Nature abstracts of fewer than thirty words were rejected. The test sits inside the per-journal quota loop, so a rejected record is replaced by the next eligible one and the equal split is preserved.
2. **At index construction, by identifier.** One OpenAlex record, `W32807916`, was excluded by identifier. Manual inspection showed it to be a table of contents misclassified as an abstract, consisting largely of dot-leader strings.
3. **At index construction, by passage.** A passage was discarded if it contained fewer than fifteen words, or if letters and whitespace accounted for less than sixty percent of its characters.

Realized effect: no record was removed by the date filter at index construction, one Nature record was excluded by identifier, and no passage was removed by the passage-level gate.

No deduplication step was applied: each source returns a unique identifier per record, and article identifiers were verified to be distinct within each source.